Supporting_information_Fish_and_Cephalopods_under_Climate_Change

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The data and analysis are presented in this document following the Material and Methods sections from the paper.

```
library(FD)
```

```
## Le chargement a nécessité le package : ade4

## Warning: le package 'ade4' a été compilé avec la version R 4.4.3

## Le chargement a nécessité le package : ape

## Warning: le package 'ape' a été compilé avec la version R 4.4.2

## Le chargement a nécessité le package : geometry

## Warning: le package 'geometry' a été compilé avec la version R 4.4.3

## Le chargement a nécessité le package : vegan

## Le chargement a nécessité le package : permute

## Warning: le package 'permute' a été compilé avec la version R 4.4.3
```

```
library(ggplot2)
```

```
## Warning: le package 'ggplot2' a été compilé avec la version R 4.4.3
```

```
library(sf)
```

```
## Linking to GEOS 3.12.1, GDAL 3.8.4, PROJ 9.3.1; sf_use_s2() is TRUE
```

```
library(raster)
```

```
## Warning: le package 'raster' a été compilé avec la version R 4.4.3

## Le chargement a nécessité le package : sp
```

```r
library(readr)
```

```
## Warning: le package 'readr' a été compilé avec la version R 4.4.3
```

2.1. Ecological traits data

The trait database corresponds for fish and cephelopods to the traits by species extracted from Aquamaps for habitat traits and Fishbase and Sealifebase and papers for cephalopods (see paper for the detailed list of references). They include traits that were available for more than 75% of the species included in the analysis. When one trait was only missing for less than 25% of the species, the value was estimated based on these decision trees, for fish (A); for cephalopods (B).

(A)

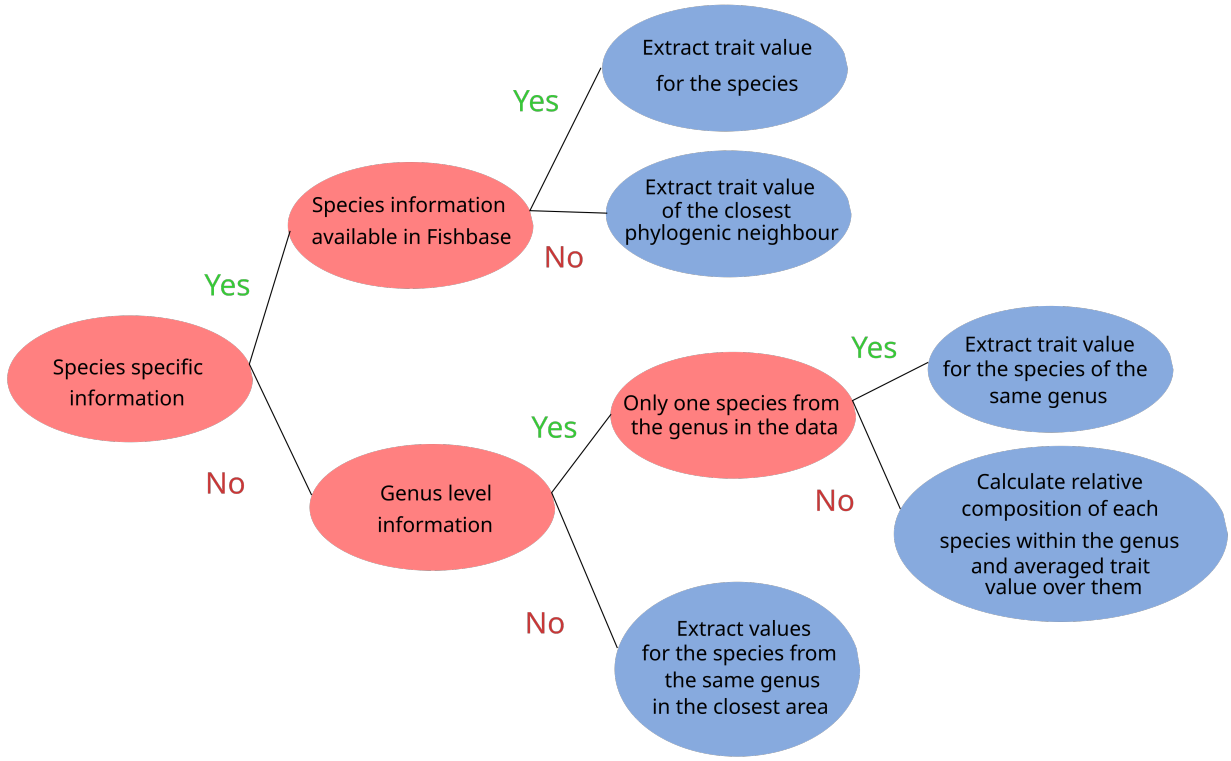


Figure 1: A caption

Open the correpsonding database

```r
BDT_FISH <- read.csv("DATA/BDT_FISH.csv")
BDT_CEPH<- read.csv("DATA/BDT_CEPHALO.csv")
```

Correlation matrix for the selected traits

(B)

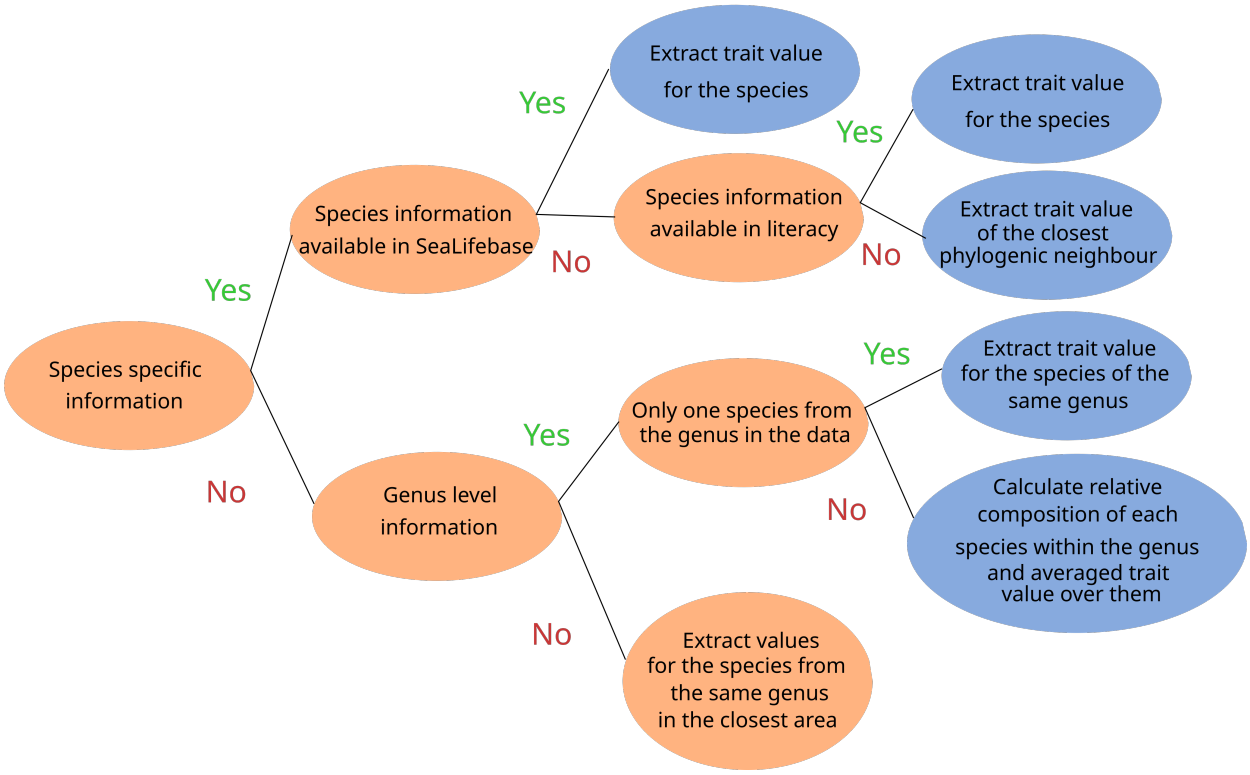
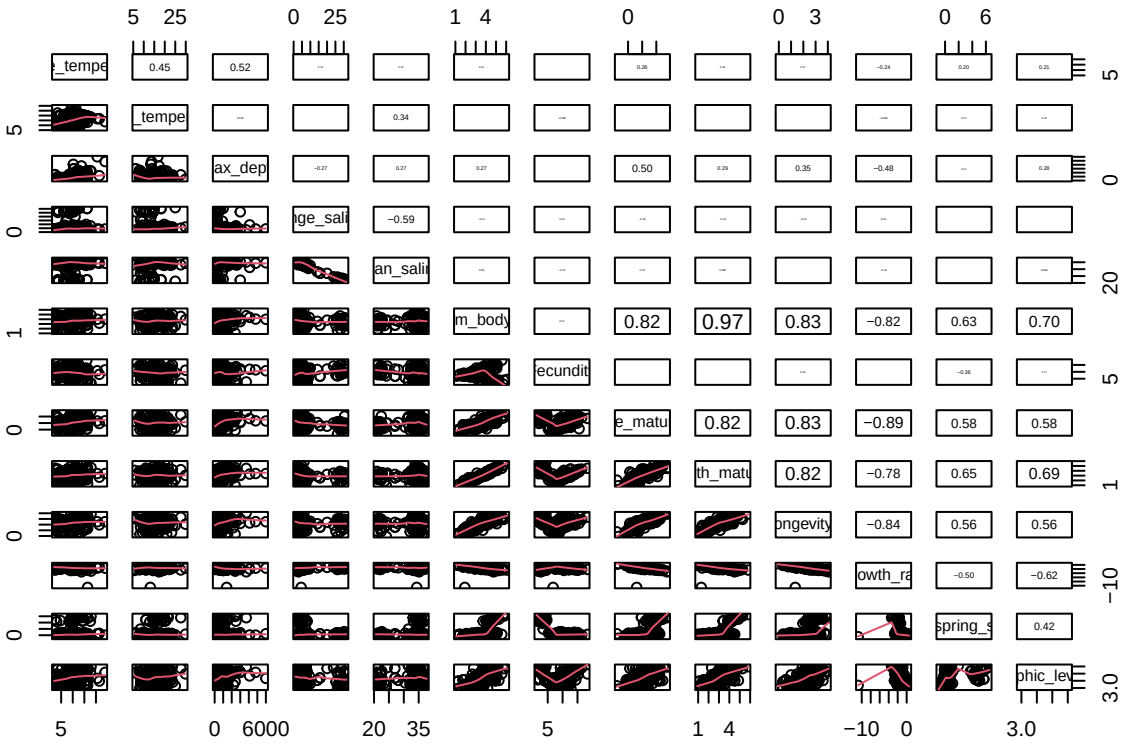


Figure 2: A caption

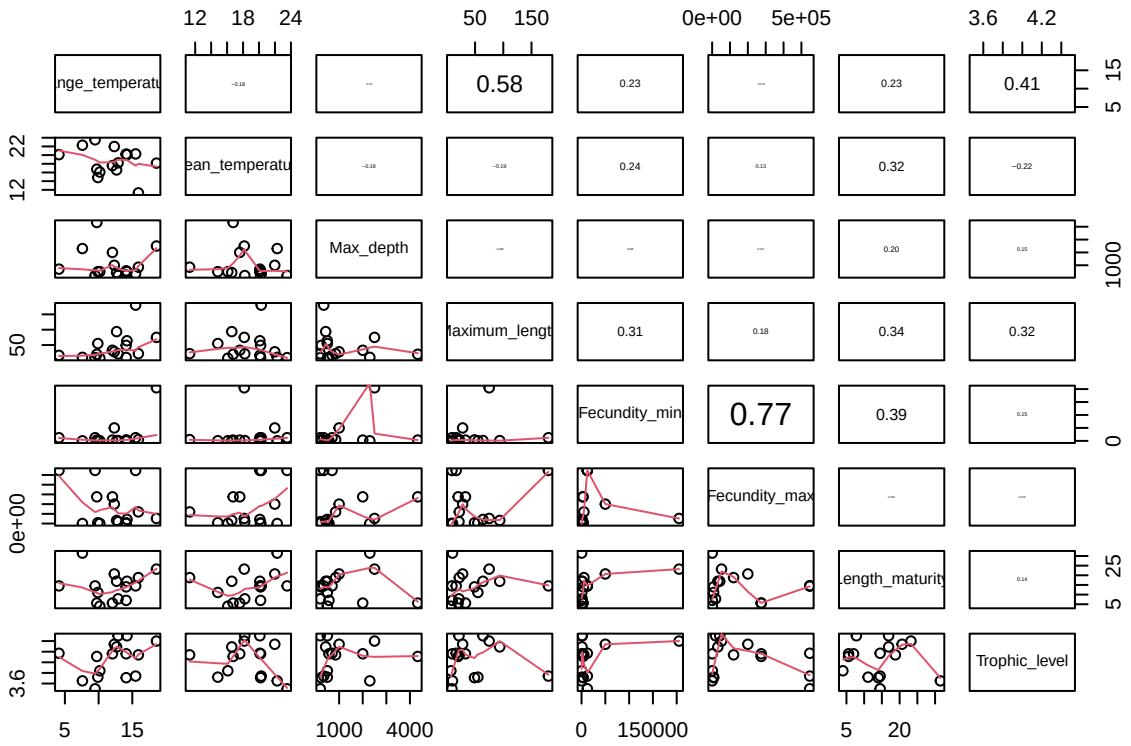
```
panel.cor.m <- function(x, y, digits=2)
{
  usrold <- par()$usr; on.exit(par("usr"=usrold))
  par("usr" = c(0, 1, 0, 1))
  r <- cor(x, y,method = "spearman", use = "complete.obs")
  r2 <- abs(cor(x, y,method = "spearman", use = "complete.obs"))
  txt <- format(c(r, 0.123456789), digits=digits)[1]
  cex <- 0.8/strwidth(txt)
  text(0.5, 0.5, txt, cex = cex * r2)

}


pairs(BDT_FISH[,2:ncol(BDT_FISH)], lower.panel=panel.smooth,
      upper.panel=panel.cor.m)
```




```
pairs(BDT_CEPH[,2:ncol(BDT_CEPH)], lower.panel=panel.smooth,
      upper.panel=panel.cor.m)
```



2.2 Species abundance data

```
load("DATA/abundance_station_species_year_fish.Rdata")
fish_abundance<-stock_list_abundance
load("DATA/abundance_station_species_year_ceph.Rdata")
ceph_abundance<-stock_list_abundance
```

2.3 Environmental data

With environmental variables followed by a x being the annual estimates and environmental variables followed by a y being the 'short term'/'immediate snapshot' of the environmental variables at the time of the sampling (november-december)

```
load("DATA/dataframe_model_2.Rdata")
```

2.4. Analysis

2.4.1. Ecological traits Community Weighted Means (CWMs)

Calculate CWM for fish

```r
# keep only selected traits and homogenized species names
keepTraits <- c("Mean_temperature","Max_depth","Range_salinity","Maximum_body_length","Growth_rate","Fe
BDT_FISH$SPECIES<-gsub("_", ".", BDT_FISH$SPECIES)
rownames(BDT_FISH)<-BDT_FISH$SPECIES
trait<-BDT_FISH
trait <- trait[,keepTraits]
abu<-vector("list", length=24)
fishFD<-vector("list", length=24)

# create empty vector for all traits we are interested in
Mean_temperature1<-Max_Depth1<-Maximum_body_length1<-Fecundity1<-Offspring_size1<-Trophic_level_1<-stati
Mean_temperature<-Max_Depth<-Maximum_body_length<-Fecundity<-Offspring_size<-Trophic_level_<-stations<-I

# loop over the years (i)
years<-c(seq(1997,2016),seq(2018,2021))
for (i in 1:24){

  abu1<-fish_abundance[[i]]
  species<-rownames(abu1)
  species<- gsub(" ", ".", species)
  rownames(abu1)<-species
  abu1<-abu1[, colSums(abu1 != 0) > 0]

  abu2<-as.data.frame(t(abu1))

  abu2<-abu2[, colSums(abu2 != 0) > 0]
  abu[[i]]<-abu2


  mtaxa <- match(colnames(abu[[i]]), row.names(trait))

  #check if there all species are matched: must be TRUE
  all(!is.na(mtaxa))

  trait1 <- trait[mtaxa,]
  all(colnames(abu[[i]])==row.names(trait1))

  fishFD[[i]]<-dbFD(trait1, abu[[i]], calc.FDiv = F, calc.FRic = F)


  Mean_temperature<-fishFD[[i]]$CWM$Mean_temperature
  Max_Depth<-fishFD[[i]]$CWM$Max_depth
  Maximum_body_length<-fishFD[[i]]$CWM$Maximum_body_length
  Fecundity<-fishFD[[i]]$CWM$Fecundity
  Offspring_size<-fishFD[[i]]$CWM$Offspring_size
  Trophic_level_<-fishFD[[i]]$CWM$Trophic_level_
  Range_salinity<-fishFD[[i]]$CWM$Range_salinity
  Growth_rate<-fishFD[[i]]$CWM$Growth_rate
  Age_maturity<-fishFD[[i]]$CWM$Age_maturity

  stations<-rownames(fishFD[[i]]$CWM)
  years2<-rep(years[[i]],length(stations))
```

```
  Mean_temperature1<-c(Mean_temperature,Mean_temperature1)
  Max_Depth1<-c(Max_Depth,Max_Depth1)
  Maximum_body_length1<-c(Maximum_body_length,Maximum_body_length1)
  Fecundity1<-c(Fecundity,Fecundity1)
  Offspring_size1<-c(Offspring_size,Offspring_size1)
  Trophic_level_1<-c(Trophic_level_,Trophic_level_1)
  Range_salinity1<-c(Range_salinity,Range_salinity1)
  Growth_rate1<-c(Growth_rate,Growth_rate1)
  Age_maturity1<-c(Age_maturity,Age_maturity1)
  station1<-c(stations,station1)
  years1<-c(years2,years1)

}
```

```
## FEVe: Could not be calculated for communities with <3 functionally singular species.
## FDis: Equals 0 in communities with only one functionally singular species.
```

```
data_for_model<-data.frame(Mean_temperature1=Mean_temperature1,
                           Max_Depth1=Max_Depth1,
                           Maximum_body_length1=Maximum_body_length1,
                           Fecundity1=Fecundity1,
                           Offspring_size1=Offspring_size1,
                           Trophic_level_1=Trophic_level_1,
                           Range_salinity1=Range_salinity1,
                           Growth_rate1=Growth_rate1,
                           Age_maturity1=Age_maturity1,
                           station=station1,
                           years1=years1)

stations<-unique(data_for_model$station1)
stations_2<-slope1<-pvalue1<-NULL

save(data_for_model, file="DATA/data_for_model_fish.Rdata")
```

Calculate CWM for cephalopods

```
BDT_CEPH$SPECIES<-gsub(" ", ".", BDT_CEPH$SPECIES)
rownames(BDT_CEPH)<-BDT_CEPH$SPECIES
trait<-BDT_CEPH[,2:ncol(BDT_CEPH)]

abu<-vector("list", length=24)
cephFD<-vector("list", length=24)

Mean_temperature1<-Range_temperature1<-Max_Depth1<-Maximum_body_length1<-Fecundity1<-Trophic_level_1<-s
Mean_temperature<-Range_temperature<-Max_Depth<-Maximum_body_length<-Fecundity<-Trophic_level_<-stations


years<-c(seq(1997,2016),seq(2018,2021))
for (i in 1:24){

  abu1<-ceph_abundance[[i]]
  species<-rownames(abu1)
```

```
  species<- gsub(" ", ".", species)
  rownames(abu1)<-species
  abu1<-abu1[, colSums(abu1 != 0) > 0]

  abu2<-as.data.frame(t(abu1))

  abu2<-abu2[, colSums(abu2 != 0) > 0]
  abu[[i]]<-abu2


  mtaxa <- match(colnames(abu[[i]]), row.names(trait))

  #check if there all species are matched: must be TRUE
  all(!is.na(mtaxa))

  trait1 <- trait[mtaxa,]
  all(colnames(abu[[i]])==row.names(trait1))

  cephFD[[i]]<-dbFD(trait1, abu[[i]], calc.FDiv = F, calc.FRic = F)

  Mean_temperature<-cephFD[[i]]$CWM$Mean_temperature
  Range_temperature<-cephFD[[i]]$CWM$Range_temperature
  Max_Depth<-cephFD[[i]]$CWM$Max_depth
  Maximum_body_length<-cephFD[[i]]$CWM$Maximum_length
  Fecundity<-cephFD[[i]]$CWM$Fecundity_max
  Trophic_level_<-cephFD[[i]]$CWM$Trophic_level
  Length_maturity<-cephFD[[i]]$CWM$Length_maturity
  stations<-rownames(cephFD[[i]]$CWM)
  years2<-rep(years[[i]],length(stations))


  Mean_temperature1<-c(Mean_temperature,Mean_temperature1)
  Range_temperature1<-c(Range_temperature,Range_temperature1)
  Max_Depth1<-c(Max_Depth,Max_Depth1)
  Maximum_body_length1<-c(Maximum_body_length,Maximum_body_length1)
  Fecundity1<-c(Fecundity,Fecundity1)
  Trophic_level_1<-c(Trophic_level_,Trophic_level_1)
  Length_maturity1<-c(Length_maturity,Length_maturity1)
  station1<-c(stations,station1)
  years1<-c(years2,years1)

}


data_for_model<-data.frame(Mean_temperature1=Mean_temperature1,
                           Range_temperature1=Range_temperature1,
                           Max_Depth1=Max_Depth1,
                           Maximum_body_length1=Maximum_body_length1,
                           Fecundity1=Fecundity1,
                           Trophic_level_1=Trophic_level_1,
                           Length_maturity1=Length_maturity1,
                           station=station1,
                           years1=years1)
```

```r
stations<-unique(data_for_model$station1)
stations_2<-slope1<-pvalue1<-NULL


save(data_for_model, file="DATA/data_for_model_cephalo.Rdata")
```